Round 10b — An exact contraction-order search for the young third-cumulant tier

Fixed bias-free ReLU MLP · n = 1024 · depth 16 · weights N(0, 2/n) · cost unit 1 u = 2^{31} FLOPs = one dense 1024^3 product ($2n^3$).

1. The young tier as a tensor network with hyperedges

For a source born at ReLU layer b and evaluated at target l > b, age r = l − b ≤ 4, the carried tensors are:

tensor	shape	role
P = P(b→l)	n × n	propagator; M_t = W_t · diag(Φ_t)
a_b	n	birth vector: diag(Φ_b) · C_off
w_b	n	birth-weight vector
A = A(b→l)	n × n	P · diag(a_b), A_ij = P_ij a_b,j
D21_l^(b)	n × n	(2,1) slice; d3 is the diagonal after summing births

Closure contribution per source:

$$D21_l^{(b)} = [\ (P\circ P)\cdot diag(w_b)\cdot A^{T} + 2(P\circ A)\cdot diag(w_b)\cdot P^{T}\ ] / 3$$

Writing entries gives:

$$\text{Term 1: } [(P\circ P)\cdot diag(w)\cdot A^{T}]_(i,k) = \Sigma_j\ w_j\ a_j \cdot P_ij^2\ P_kj$$

$$\text{Term 2: } [(P\circ A)\cdot diag(w)\cdot P^{T}]_(i,k) = \Sigma_j\ w_j\ a_j \cdot P_ij^3\ P_kj$$

Both terms share the trailing factor P_kj, hence the exact collapse:

$$D21_l^{(b)}_(i,k) = (1/3)\ \Sigma_j\ g_(i,j)^{(b,l)} \cdot P_(b\to l)_(k,j),\ g_(i,j) = w_j\ a_(b,j) \cdot P_ij^2(1 + 2P_ij)$$

Equivalently, D21^(b) = (1/3) G^(b) P^{T}. The supplied n=2 hand check gives term-1 = 7 and term-2 = 26 for entry (1,1), matching the collapsed form.

2. Exact FLOP count at the real shapes

step	operation	FLOPs	u
1. transport P	M · P	$2n^3$ = 2,147,483,648	1.000
2. build G	entrywise	≈ $6n^2$ = 6,291,456	0.003
3. contract	G · P^{T}	$2n^3$ = 2,147,483,648	1.000
total / pair	collapsed		2.003

There are Σ_(r=1..4)(16−r) = 58 possible young (birth,target) pairs. Using the ledger's stated effective count of 54 pairs:

$$54 \times 2.003\ u = 108.2\ u$$

This reproduces approximately 54 u transport + 50 u D21. The supplied text's phrase “four dense products per pair” is inconsistent with the 104 u ledger unless A transport is free; the collapse makes A unnecessary.

3. Best orders found

Order A — collapsed forward (recommended). Keep P_(b→l), advance P ← M·P, never form A, build G entrywise, then compute G·P^{T}. Exact cost: 108.2 u for 54 effective pairs.

Order B — forward ages 1–2, reverse ages 3–4. Reverse mode carries both query legs and still requires the same dense contraction. Cost remains 2.003 u/pair; no saving.

Order C — precomputed consecutive maps. Precomputing Q_t = M_tM_(t−1) costs about 15 u. Some transport is reduced, but ages 3–4 still need additional products and the Q_t are not reusable across all birth/target offsets. It does not beat the collapsed order under the stated ledger.

Order D — rank/sharing check. G^(b,l) differs by birth-specific w_b, a_b and P_(b→l), so no exact common factor exists across births. Dense full-rank contractions remain $2n^3$.

4. Verdict

No mixed or hierarchical contraction order beats the current cost by 10% under the stated exact dense model; the ≤55 u target is not attainable by reordering alone.

The per-pair lower-bound structure is one dense transport (1.000 u) + one dense contraction (1.000 u) + $O(n^2)$ entrywise work. Over 54 effective pairs this is about 108 u.

Why the floor is pinned: distinct births have non-nested P; G is birth-dependent; batching is FLOP-equivalent; reverse mode is an isomorphic contraction order.

Real saving already realized: never form A. Further exact gains are primarily memory-side (incremental accumulation), not another contraction-order trick.

5. Verification op list

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for each target l:
  for b in {l−4, l−3, l−2, l−1}:
    P = P_(b→l)
    G = w_b[None,:] · a_b[None,:] · P² · (1 + 2P)
    D21_l += G @ P.T / 3.0
  d3 = diag(D21_l)
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Identity to verify against the uncollapsed reference: $D21^{(b)}_{(i,k)} = (1/3)\ \Sigma_j\ w_j\ a_j\ (P_{ij}^2 + 2P_{ij}^3)\ P_{kj}$.

6. Provenance

This PDF is a clean typeset rendering of the supplied HTML content. No new numerical experiment was run during conversion. The arithmetic shown uses the source convention $2n^3 = 2^{31}$ FLOPs = 1 u.